

Replication README

Data and Code for: Equilibrium in the Market for Public School Teachers: District Wage Strategies and Teacher Comparative Advantage

Barbara Biasi, Chao Fu, and John Stromme

This README documents the replication package for the Journal of Political Economy paper “Equilibrium in the Market for Public School Teachers: District Wage Strategies and Teacher Comparative Advantage.”

Overview

The code in this replication package constructs teacher-, student-, district-, model-estimation, and simulation datasets and generates the tables and figures listed below and displayed in the paper and Online Appendix.

The original analysis relies on confidential Wisconsin Department of Public Instruction (WDPI) data and cannot be redistributed in this public archive. The public replication package hence includes fixed synthetic input data in `synthetic_data/raw/`. These files are fictitious data created to match the file names, folder structure, variable names, storage formats, and key relationships needed by the replication code. They are not sampled from, perturbed from, or otherwise derived from confidential WDPI records. Results produced from these synthetic data verify the code path and file dependencies but do not reproduce the published numerical estimates.

Public replicators should run `code/master_synth.do`. It reads the synthetic raw inputs from `synthetic_data/` and the public proficiency input from `data/public/`, constructs intermediate/model/simulation files locally, and writes synthetic-run exhibits to `out_synth/`. The confidential-data driver is `code/master.do`; it is not needed for the public synthetic-data replication. In this package, `out_synth/` is the output-simulated folder: it contains the expected tables and figures from the public simulated-data run. `out/` is reserved for actual restricted-data outputs from `code/master.do` and will not be generated in the public replication.

Data Availability and Provenance Statements

Statement About Rights

The confidential WDPI staff, roster, demographic, and test-score data used in the paper cover 2006–2016. The authors first obtained access in November 2017 under a Data User Agreement with WDPI. WDPI did not provide a formal version number, extract identifier, or release number for the delivery. These restricted data are not distributed with this replication package.

Researchers who want access to the confidential data can submit a request to WDPI at:

<https://dpi.wi.gov/wise/data-requests/confidential-data-application>

There is no application or access fee. Access requires IRB approval and a signed Data User Agreement, is subject to WDPI review and approval, and may take several months. The Data User Agreement requires the confidential data to be destroyed after the project is completed. Redistribution of confidential WDPI microdata is not permitted.

In June 2026, the JPE Data Editor approved the use of synthetic data for the public replication in place of the confidential WDPI files.

`real_data/` contains shareable auxiliary inputs that are either constructed by the authors or publicly available. These files are included in the public replication archive subject to the applicable source terms and agreements. `data/public/` contains an unmodified public ED Data Express district-level mathematics assessment file, its provider README, and its provider data notes. These files support the calculation of the district proficiency ranges discussed in the paper. `synthetic_data/raw/` contains the fictitious WDPI-style inputs distributed for the public replication.

Summary of Availability

Data source	Location expected by code	Publicly provided?	Notes
WDPI staff, roster, and student test-score files	File names listed below	No	Confidential/restricted data covering 2006–2016. First accessed in November 2017; no formal version or release number was provided. Access is free but requires IRB approval, a signed DUA, and WDPI approval, which may take several months.
Auxiliary district crosswalks, distance files, shapefiles, elections, and NCES/ACS extracts	<code>real_data/</code>	Yes	Shareable files constructed by the authors or publicly available.
District mathematics proficiency data	<code>data/public/proficiency_data/SY2021_FS175_DG583_LEA_data_files/</code>	Yes	Unmodified public 2020–2021 LEA-level mathematics assessment data from the U.S. Department of Education’s ED Data Express Data Library (File Specification 175, Data Group 583), plus the provider README and data notes. Used by <code>code/descriptives/intext_numbers.do</code> .

Data source	Location expected by code	Publicly provided?	Notes
Synthetic public replication data	synthetic_data/raw/	Yes	Fixed fictitious WDPI-style raw files distributed with the public package. Together with the included auxiliary inputs in real_data/ and public proficiency input in data/public/, these are sufficient to run code/master_synth.do. The simulation script is also included at synthetic_data/code/make_simulated_data.do so public replicators can inspect or regenerate the fictitious inputs if needed.
Generated intermediate/model/simulation data	intermediate_data/, model_data/, simulation_data/	No	Created locally when code/master_synth.do runs. They should not be treated as source data.
Expected simulated-run outputs	out_synth/	Yes	Tables and figures generated from the simulated inputs. This is the package's output-simulated folder.
Actual restricted-data outputs	out/	No	Reserved for outputs from the confidential-data workflow. Public replicators will not generate this folder from the public simulated data.

Preservation and Licensing

The final public replication package will be preserved in the JPE/Harvard Dataverse. The authors' local copies of the confidential WDPI data will be destroyed when required by the governing Data User Agreement. The underlying administrative records remain in the custody of WDPI, which will continue to accept applications from qualified researchers for restricted access for at least five years following publication. During that period, the authors will preserve the replication code, synthetic data, codebook, provenance documentation, and access instructions. The authors will also provide reasonable assistance in response to requests for clarification and replication, including assistance to approved researchers seeking to obtain a substantially similar extract and run the analysis code.

The authors' code in code/ and synthetic_data/code/ is released under the MIT License. The authors' original synthetic data and documentation are released under the Creative Commons Attribution 4.0 International License (CC BY 4.0). Third-party auxiliary inputs are not relicensed, including files under real_data/ and data/public/, and remain subject to their original source terms. See LICENSE for details.

Raw Datasets Used in the Analysis

The following list contains all the raw files used in the analysis. WDPI staff, roster, and student files are confidential and not provided. Files under real_data/ are shareable auxiliary inputs constructed by the authors or publicly available. Files under data/public/ are unmodified public source files distributed with their provider documentation.

Raw input	Source/provenance from local evidence
Linked_Staffing_Data_10202015.csv	Confidential WDPI linked staffing data.
06staff.txt through 14staff.txt	Confidential WDPI staff files.
15staff.csv, 16staff.csv	Confidential WDPI staff files.
student_demographics.csv	Confidential WDPI student demographics.
testscores.csv	Confidential WDPI student test scores.
data/public/proficiency_data/SY2021_FS175_DG583_LEA_data_files/SY2021_FS175_DG583_LEA.csv	U.S. Department of Education, ED Data Express, 2020–2021 <i>Performance on Statewide Mathematics Assessment</i> , LEA level, File Specification 175, Data Group 583, as of May 25, 2022. The file is an unmodified provider download with 1,394,390 data records.
data/public/proficiency_data/SY2021_FS175_DG583_LEA_data_files/SY2021_FS175_DG583_LEA_README.txt	Provider README supplied with the ED Data Express download.
data/public/proficiency_data/SY2021_FS175_DG583_LEA_data_files/SY2021_FS175_DG583_LEA_data_notes.csv	Provider data notes supplied with the ED Data Express download, including the provider's caution about 2020–2021 assessment participation during the COVID-19 pandemic.
real_data/cesa.dta	CESA-district crosswalk manually created by authors using WDPI CESA information .
real_data/wisconsin_elections.xlsx	Daily Kos Elections statewide election results and county-congressional-district crosswalk, accessed in 2022; see real_data/readme_elections.rtf and the formal citations below.
real_data/crosswalk_district_county.dta	District-county crosswalk prepared by authors using the Opportunity Insights county-commuting-zone-state crosswalk documented below.
real_data/crosswalk_census_state_id.csv	Census district ID to state district ID crosswalk prepared by authors using NCES EDGE School District Boundaries . The exact source vintage was not recorded.
real_data/district_county_cz.dta	District-county-commuting-zone crosswalk prepared by authors using the Opportunity Insights county-commuting-zone-state crosswalk documented below.

Raw input	Source/provenance from local evidence
real_data/district_distance_sdid.dta	District-to-district distance matrix constructed by authors using Google Maps in September 2022.
real_data/nces_acs_2013-2017/EDGE_Export_101244429626/DP03.6_001_USSchoolDistrictAll_101244423682.txt	NCES/ACS 2013–2017 school-district income extract. The original download date was not recorded.
real_data/nces_acs_2013-2017/EDGE_Export_101244639981/DP03.8_001_USSchoolDistrictAll_101244637329.txt	NCES/ACS 2013–2017 school-district poverty extract. The original download date was not recorded.
real_data/nces_acs_2013-2017/EDGE_Export_1012447445/DP02.7_001_USSchoolDistrictAll_101244740697.txt	NCES/ACS 2013–2017 school-district education extract. The original download date was not recorded.
real_data/nces_acs_2013-2017/EDGE_Export_101253245819/DP05.1_001_USSchoolDistrictAll_101253240998.txt	NCES/ACS 2013–2017 school-district age extract. The original download date was not recorded.
real_data/district_shapefiles/WI_School_Districts.*	WDPI <i>School Districts, Wisconsin</i> , edition 8.0, revised June 26, 2024 and valid for the 2024–2025 school year, plus converted Stata files. Downloaded in 2025 from the WDPI GIS Maps page.
real_data/cz_id.csv	District-commuting-zone crosswalk prepared by authors using the Opportunity Insights county-commuting-zone-state crosswalk documented below.
real_data/district_id_tag.dta and real_data/district_id_tag_validation.dta	Data-district-id to model-district-id crosswalks prepared by authors.

Public District Proficiency Data Codebook

The ED Data Express file is an LEA-level extract of 2020–2021 statewide mathematics assessment performance. A row identifies a reported result for an LEA, student population or subgroup, and grade. The in-text-number script uses the rows for All Students in LEA, Grade 5, and the states California, Texas, and Wisconsin. The provider reports some percentages exactly and others as disclosure ranges or one-sided bounds. The script therefore parses the reported Value strings rather than treating every row as an exact point estimate. For every qualifying numeric report, it extracts the first numeric endpoint and, for an interval, the second numeric endpoint; an exact or one-sided report uses its stated numeric value as both endpoints. It then takes the smallest lower and largest upper endpoint by state. The resulting ranges are 5–94% in Wisconsin, 5–89% in California, and 6–90% in Texas.

Raw variable	Description
School Year	School year covered by the record; this extract reports 2020–2021.
State	State name.
NCES LEA ID	Seven-digit NCES local education agency identifier.
LEA	Local education agency name.
School	School name when applicable; blank in this LEA-level extract.
NCES SCH ID	NCES school identifier when applicable; blank in this LEA-level extract.
Data Group	EDFacts data-group identifier; 583 denotes the assessment-performance group used here.
Data Description	Provider description of the reported measure.
Value	Reported percentage or provider code. Numeric percentages can be exact (for example, 42%), interval-censored (for example, 40–44%), or one-sided (for example, <=10% or >=90%); suppression and missing-value codes can also occur.
Numerator	Numerator when released by the provider; not populated for the proficiency-percentage records used here.
Denominator	Number of students underlying the reported percentage.
Population	Population represented by the record; All Students for the rows used here.
Subgroup	Reporting subgroup; the calculation keeps All Students in LEA.
Characteristics	Additional subgroup characteristic when applicable.
Age/Grade	Student age or grade; the calculation keeps Grade 5.
Academic Subject	Assessed subject; Mathematics in this extract.
Outcome	Reported outcome; Percent Proficient in this extract.
Program Type	Program classification when applicable.

The provider notes that 2020–2021 assessment participation varied because of the COVID-19 pandemic; results may not be representative and should not be assumed comparable with other years. See SY2021_FS175_DG583_LEA_data_notes.csv for the full provider note.

Restricted WDPI Data Codebook

The confidential WDPI inputs are not distributed, but the following codebook documents the variables obtained from WDPI and expected by the build code. It is intended to let an approved researcher determine whether a newly obtained extract is substantially similar. Variable names shown in parentheses are the canonical lower-case names used by the Stata build after import. Blank fields, provider missing-value codes, and NA should be treated as missing unless a more specific rule is stated below.

Annual staff files

The unit of observation in 06staff.txt through 14staff.txt and 15staff.csv through 16staff.csv is a staff member–district–school– position–assignment–year record. A staff member can therefore have multiple records in a year. The legacy files are fixed-width; their column positions are specified in code/build/staff_build.do. The 2015–2016 files use the raw CSV headers shown below. The file year supplies the analysis year.

Raw/canonical variable	Description and coding expected by the build
IDNbr (id)	Provider staff identifier. Numeric in the annual files; used with names and demographics to construct longitudinal links.
LastName, FirstName (lastname, firstname)	Staff surname and given name used only for record linkage and name reconciliation. String; confidential direct identifiers.
FileNumber (filenumber)	Longitudinal staff file number, available in the later files and recovered from the linked-staffing file when necessary. String or numeric identifier; leading zeros may occur.
Gndr (gender)	Provider gender code. The analysis recognizes F and M; blank or NA is missing.
RaceEthn (race)	Provider race/ethnicity code. The analysis uses B for Black, H for Hispanic, and A for Asian; other valid provider codes are retained until the analysis-specific indicators are created.
BirthYear (birth)	Four-digit year of birth used for linkage and experience-consistency checks.
HighDegree (highedu)	Highest-degree code. Codes used by the analysis are 4 bachelor's, 5 master's, 6 specialist, and 7 doctorate.
YearSession	Calendar/session year in the CSV files. The build assigns year from the annual filename so that all annual formats are handled consistently.
CntrctDays (dayscontract)	Number of contract days for the staff assignment. Records with zero or missing contract days are excluded.
LocalExp (localexp)	Provider-reported years of experience in the local agency. Numeric, with missing values allowed.
TotalExp (totalexp)	Provider-reported total years of experience. Numeric, with missing values allowed.
TotSalary (salary)	Total annual salary associated with the record, in nominal dollars as supplied. Zero and missing values are excluded.
TotFringe (fringe)	Total fringe benefits associated with the record, in nominal dollars as supplied.
StaffCat (staffcat)	Provider staff-category code.
HireAgncyCd, WorkAgncyCd (hireagencycode, workagencycode)	Official codes for the hiring and work agencies. The work-agency code becomes the analysis district identifier.
HireAgncyTyp, WorkAgncyTyp (hireagencytype, workagencytype)	Provider hiring- and work-agency-type codes. The teacher build retains district- and school-employee types 3 and 4.
SchoolCd (schoolcode)	Official school code within the work agency. Numeric after import; missing school codes are excluded from the teacher panel.
PositionCd (position)	Provider position code. Codes used to define the teacher sample are 53 for teacher and 43 for short-term substitute teacher.
AssgnAreaCd (area)	Provider assignment-area/subject code. The build uses these codes to create mathematics, English, reading, art, science, Title I, elementary, and cross-categorical assignment indicators.
LowGrd, HighGrd (lowgrade, highgrade)	Lowest and highest grade served, stored as provider text codes.
LGSortCd, HGSortCd (lowgradecode, highgradecode)	Numeric sort codes corresponding to the lowest and highest grades served.
Bilingual (bilingual)	Provider bilingual-assignment indicator, normally Y/N or blank.
AssgnFTE (fte)	Assignment full-time-equivalent percentage; 100 represents one full-time assignment. Zero and missing FTE records are excluded.
WorkLocationName (workagencyname)	Name of the work agency or district. String.
SchoolName (schoolname)	School name. String.
GrdLevel (gradelevel)	Provider school/assignment grade-level code.
CESANumber (cesa)	Cooperative Educational Service Agency code. String or numeric provider code.
CntyNbr (county)	Provider county code.
CntyName	Provider county name. Present in the CSV files but not retained by the analysis build.
SchoolMailingAddress1, SchoolMailingAddress2, SchoolShippingAddress1, SchoolShippingAddress2, MailCity, MailSt, ShipCity, ShipSt, ShipZipCd	Work-location mailing and shipping address components. Present in the CSV files but not retained by the analysis build.
MailZipCd (zipcode)	Work-location mailing ZIP code. String so leading zeros are preserved.
Phone, AdminName, FormerLastNm	Work-location phone, administrator name, and former staff surname. Present in the CSV files but not retained by the analysis build.

Raw/canonical variable	Description and coding expected by the build
LTSUB (lt_substitute)	Long-term-substitute indicator; Y denotes a long-term substitute.
SubCntrctd (subcontracted)	Subcontracted-employee indicator; records coded Y are excluded from the teacher panel.

The address, telephone, administrator, county-name, and former-name variables are documented for completeness but are not retained by `staff_build.do` and are not required in a substantially similar analysis extract.

Linked staffing file

`Linked_Staffing_Data_10202015.csv` is a staff-year/worksite linkage file used to recover longitudinal file numbers before the annual staff files are combined.

Raw variable	Description and coding expected by the build
last_name, first_name	Staff surname and given name used for linkage. String; confidential direct identifiers.
file_num	Longitudinal staff file number. Converted to numeric when possible; zero, NA, and nonnumeric values are treated as missing.
year	Four-digit calendar/session year. The build retains 2006 and later.
race	Provider race/ethnicity code; NA is treated as missing.
birth_year	Four-digit year of birth used for linkage.
sex	Provider sex/gender code. The build recognizes F and M.
distidworksite	Official district/worksite identifier, converted to numeric when possible.

Student-demographic file

The unit of observation in `student_demographics.csv` is a student-school year. The expected key is `STUDENT_KEY` and `SCHOOL_YEAR`.

Raw variable	Description and coding expected by the build
SCHOOL_YEAR	School-year string in YYYY–YYYY form. The first year is used as the initial analysis year.
DIST_ACCTBL_CODE	Official accountable-district code.
SCH_ACCTBL_CODE	Official accountable-school code.
STUDENT_KEY	De-identified longitudinal student identifier shared with the test-score file.
GRADE	Provider grade code. The test-score analysis uses grades 3–8.
GENDER	Provider gender code. F creates the female indicator; blank is missing.
RACE_ETHNICITY	Provider race/ethnicity label. Labels used by the analysis include Black, Hispanic, Asian, and Amer Indian; other provider categories may be present.
DISAB_STATUS	Disability-status indicator or provider status code; numeric/blank values are retained and carried across student years when missing.
EL_STATUS	English-learner-status indicator or provider status code; numeric/blank values are retained and carried across student years when missing.
ECON_DISADVANTAGE	Economic-disadvantage indicator or provider status code; numeric/blank values are retained and carried across student years when missing.

Student test-score file

The unit of observation in `testscores.csv` is a student–subject–school-year test record. The expected key is `STUDENT_KEY`, `SCHOOL_YEAR`, and `TEST_SUBJECT`, allowing one record per tested subject in a student-year.

Raw variable	Description and coding expected by the build
SCHOOL_YEAR	School-year string in YYYY–YYYY form. The build uses both the first and second year to align fall and spring tests.
DIST_ACCTBL_CODE	Official accountable-district code; renamed <code>district_code</code> .
SCH_ACCTBL_CODE	Official accountable-school code. The build extracts the numeric school code from this string.
STUDENT_KEY	De-identified longitudinal student identifier shared with the demographic file.
GRADE	Tested grade. The analysis retains grades 3–8 and excludes grade 10.
TEST_NAME	Provider assessment/program name. String.
TEST_SUBJECT	Tested subject. The build expects Reading/Mathematics in earlier school years and ELA/Mathematics in later school years, according to the year alignment implemented in <code>testscores_build.do</code> .
TEST_SCALED_SCORE	Numeric provider scaled score. The build standardizes this within grade, year, and subject.
TEST_STDERR	Numeric standard error supplied with the test score. It is imported but not used in the final standardized-score panel.
TEST_PROF_LVL	Provider proficiency-level label, such as basic, proficient, or advanced categories. It is imported but not used in the final standardized-score panel.

The public `synthetic_data/raw/staff/temp.dta` and `synthetic_data/raw/testscores_indlev/dem.dta` files are validation/support copies, not additional confidential source datasets. The production build constructs its working demographic file from `student_demographics.csv`.

Synthetic Public Data

The public archive replaces the confidential WDPI staff and student inputs with fixed synthetic files under `synthetic_data/raw/`. These files are fictitious records with the same broad structure expected by the build scripts.

Synthetic input	Unit of observation	Notes
<code>synthetic_data/raw/staff/06staff.txt</code> through <code>14staff.txt</code>	Staff district-school-position-year	Fixed-width fictitious staff files matching the legacy WDPI staff-file layout.
<code>synthetic_data/raw/staff/15staff.csv</code> , <code>16staff.csv</code>	Staff district-school-position-year	CSV fictitious staff files matching the later WDPI staff-file layout.
<code>synthetic_data/raw/staff/temp.dta</code>	Staff-year linkage support	Fictitious support copy checked by the synthetic-data validator; it is not read by the production synthetic master.
<code>synthetic_data/raw/teacher_data_linked/Linked_Staffing_Data_10202015.csv</code>	Staff-year linkage support	Fictitious linked-staffing support file required by <code>oldteacher_roster_build.do</code> .
<code>synthetic_data/raw/testscores_indlev/student_demographics.csv</code>	Student-year	Fictitious student demographics.
<code>synthetic_data/raw/testscores_indlev/dem.dta</code>	Student-year	Stata-format support copy checked by the synthetic-data validator; the production build creates its working demographics file from the CSV.
<code>synthetic_data/raw/testscores_indlev/testscores.csv</code>	Student-subject-year	Fictitious math, reading, and ELA test-score records.

The synthetic staff files are designed so staff IDs overlap heavily across years, staff observations are district-school-position-year records, most person-years have one district-school-position, and most one-position person-years are teachers. Synthetic test-score and demographic files share fictitious student IDs and school years, with student records repeated across years. The synthetic data also include nonempty cells needed by value-added, teacher-choice, event-time, subgroup, wage-schedule, and map/figure code.

The synthetic inputs are distributed as final fictitious raw files. The simulation script used to generate them is included for transparency at `synthetic_data/code/make_simulated_data.do`. That script creates the fictitious WDPI-style raw inputs from scratch, does not read confidential records, calls `synthetic_data/code/02_make_synthetic_data.do`, and then runs `synthetic_data/code/03_validate_synthetic_structure.do` to check the simulated data's structure and key relationships. Running it overwrites files under `synthetic_data/raw/` and refreshes the public manifest/notes. Additional documentation is in `synthetic_data/documentation/synthetic_data_notes.md` and `synthetic_data/documentation/synthetic_data_manifest.csv`; both files are included in the public archive.

Repository Structure

The main folders and files in the replication folder are:

Path	Purpose
<code>README.md</code>	This replication guide.
<code>LICENSE</code>	License terms for the authors' code, synthetic data, and documentation, with third-party materials excluded from the license grant.
<code>code/</code>	Stata, R, and Fortran code for data construction, analysis, model, and exhibit generation.
<code>code/master_synth.do</code>	Public Stata driver using the fixed synthetic WDPI-style inputs and writing exhibits to <code>out_synth/</code> .
<code>code/master.do</code>	Main Stata driver. Defines globals and calls build/exhibit scripts.
<code>code/produce_exhibits.do</code>	Secondary Stata driver. After the master and Fortran results are produced, calls exhibit scripts to complete the remaining outputs.
<code>code/build/</code>	Stata scripts that construct roster, staff, teacher panel, test-score, value-added, wage-schedule, model-estimation, and simulation input files.
<code>code/descriptives/</code>	Stata scripts for descriptive tables, figures, appendix robustness tables, and map/figure construction.
<code>code/auxiliary_regressions/</code>	Stata scripts for auxiliary teacher-choice and wage regressions and robustness variants.
<code>code/fortran_model/</code>	Fortran code to run the model and all counterfactual and other modeling analyses.
<code>code/fortran_model/common_f90s/</code>	Shared Fortran modules, stored once; the <code>entry/</code> branch is used by <code>entry/forward</code> scenarios and <code>non_entry/</code> by all other scenarios.
<code>code/fortran_model/run_all_local.sh</code>	Portable compiler/runner for all 30 Fortran scenarios on macOS and Unix-like systems. It compiles only unless the replicator explicitly supplies <code>--run</code> .
<code>code/fortran_model/example_slurm_batchfile</code>	Common Slurm submission script used for the submitted scenario runs; its site-specific settings must be adapted before use on another cluster.
<code>original_fortran_instruction_files/</code>	Archived full-run <code>instruction.txt</code> files for all 30 Fortran scenarios, with paths matching the active scenario directories.

Path	Purpose
code/Figure_B9/	Figure B9 Stata code and R simulation helper for the identification illustration.
data/public/	Unmodified third-party public source data used to calculate in-text statistics, with provider documentation.
intermediate_data/	Working intermediate Stata datasets. Current files may be derived from restricted inputs.
model_data/	Model-estimation inputs and CSV exports generated by <code>build_model_dataset.do</code> and related scripts. Current files may be derived from restricted inputs.
simulation_data/	Pre-Act 10 simulation input files generated by <code>build_model_dataset_pre2011.do</code> . Current files may be derived from restricted inputs.
out/	Actual restricted-data exhibit outputs from <code>code/master.do</code> . This folder is not generated by the public simulated-data replication.
out_synth/	Expected simulated-run outputs from <code>code/master_synth.do</code> using the public synthetic data. This is the package's output-simulated folder.
real_data/	Shareable auxiliary inputs that are either constructed by the authors or publicly available.
synthetic_data/raw/	Fixed public synthetic inputs that replace the confidential WDPI staff and student files.
synthetic_data/code/make_simulated_data.do	Public Stata script that regenerates and validates the simulated WDPI-style raw inputs from scratch.
synthetic_data/documentation/	Public manifest and notes describing the simulated raw inputs. Private profiling logs and Stage 1 planning notes are not required for the public archive.

Computational Requirements

Software

Software	Requirement/evidence
Stata 19	Stata/SE or Stata/MP can run the public synthetic workflow. The confidential driver requests a larger variable ceiling and therefore requires Stata/MP. The public workflow was tested with StataNow/MP 19.5 for Apple Silicon, 21 May 2025 revision. <code>code/master.do</code> and <code>code/master_synth.do</code> verify or install the required SSC packages before running. An Internet connection is needed on the first run unless they are already installed.
Stata packages	The tested SSC versions are <code>estout</code> 3.33, <code>esttab</code> 2.1.4, <code>ftools</code> 2.50.0, <code>reghdfe</code> 6.13.1, <code>coefplot</code> 1.8.8, <code>binscatter</code> 7.02, <code>carryforward</code> 4.5, <code>spmap</code> 1.3.2, <code>sepscatter</code> 1.1.0, <code>blindschemes</code> 1.3.0, and <code>require</code> 1.3.1. <code>blindschemes</code> provides the <code>plotplain</code> graph scheme, while <code>require</code> is used by <code>reghdfe</code> to manage dependencies. The drivers install current SSC releases rather than enforcing these exact versions.
R	Required only for <code>code/Figure_B9/cutoff_prefs_id_example_graph.R</code> . The analysis used R 4.2.3. The helper was also tested with R 4.5.1, <code>haven</code> 2.5.5, <code>extRemes</code> 2.2.1, and <code>MASS</code> 7.3-65. It creates the configured user library when needed and installs missing packages from CRAN.
Fortran and MPI	The model was originally run with GNU Fortran (GCC) 11.5.0 20240719 (Red Hat 11.5.0-14) and <code>mpiexec</code> (OpenRTE) 4.1.1. The distributed code was also compiled successfully on Apple Silicon with GNU Fortran 16.1.0 and Open MPI 5.0.9.
BLAS/LAPACK	BLAS and LAPACK are required by the Fortran model. The portable runner links Homebrew OpenBLAS on macOS and <code>-lblas -llapack</code> on other Unix-like systems. <code>FORTTRAN_LINK_FLAGS</code> can override this choice.
Shell/OS	Command-line examples use a Unix-like shell. The portable Fortran runner requires Bash. Start Stata in the repository root as instructed below. The Stata drivers locate the replication root from Stata's current working directory; no machine-specific path edits are required.

Tested Hardware, Runtime, and Storage

The public Stata workflow and local compilation checks were run on a MacBook Pro (Mac16,6) with an Apple M4 Max processor (16 CPU cores), 128 GB of memory, and macOS 26.5.2. The installed Stata/MP license used two Stata cores.

In two clean validation runs on this machine, optional synthetic-input regeneration took approximately 1 hour 31 minutes to 1 hour 43 minutes, and `code/master_synth.do` took approximately 2 hours 43 minutes to 2 hours 53 minutes. The Figure B9 R helper took less than three seconds. The complete 30-scenario Fortran stage took approximately 12 hours 29 minutes with eight MPI ranks, and the final exhibit step took less than five seconds.

Using the included synthetic raw inputs, the required Stata, R, Fortran, and final-exhibit pipeline therefore took approximately 15 hours 13 minutes. Including optional synthetic-input regeneration increased the total to approximately 16 hours 44 minutes. Replicators should allow an overnight or full-day run; hardware and MPI configuration can materially affect runtime.

The complete working tree occupied approximately 3.8 GB after the Stata workflow generated its intermediate and model data. Replicators should allow for an additional approximately 269 MB for the uncompressed public ED Data Express proficiency file and should allow at least 5 GB of free disk space. More space may be needed for Fortran logs, platform-specific build files, or additional validation copies.

Controlled Randomness

`code/master_synth.do` sets both `set seed 12345` and `set sortseed 12345` to control Stata randomness. The synthetic master also normalizes opened/closed timestamps in Stata table logs so repeated public runs produce the same substantive synthetic tables and figures. The simulated-data generator `synthetic_data/code/make_simulated_data.do` sets both `set seed`

20260613 and set sortseed 20260613. The generator also normalizes Stata .dta timestamps and unused padding in the two public support .dta files, so regenerated simulated raw inputs are byte-for-byte stable across reruns.

The Figure B9 R helper sets its own R seed.

For the Fortran code, each main.f90 sets the random-number generator to the same four fixed seed values.

Description of Programs and Execution Order

The public driver, code/master_synth.do, calls the core scripts in this order. The confidential driver, code/master.do, follows the analogous restricted-data workflow.

1. Build teacher panel without contribution measures: oldteacher_roster_build.do, staff_build.do, teachers_build.do, teachers_panel.do.
2. Build test-score and teacher-side value-added inputs: va_teacherside_build.do, testscores_build.do.
3. Estimate teacher contribution measures: valueadded_calculation.do, valueadded_calculation_classmodel.do, valueadded_calculation_stockstaiger_nocomparativeadv.do.
4. Construct final analysis panel: teachers_with_allva.do.
5. Generate model-estimation inputs: schedule_new.do, gen_omegas.do, gen_omegas_info.do, build_model_dataset.do, build_teachers_offer_set.do, auxiliary_models_teachers.do, aux_other_moments.do, auxiliary_models_districts.do, build_model_dataset_info0.do, build_model_dataset_info1.do, build_model_dataset_info2.do, build_model_dataset_info3.do, auxiliary_models_teachers_info_matches.do, omega_databuild.do, budget_overtime.do, build_distance_matrix.do, build_wageschedule_exitentry.do, fix_estsample_age.do.
6. Generate pre-Act 10 simulation inputs: build_model_dataset_pre2011.do, build_simulation_1ndist.do.
7. Generate the in-text calculations: descriptives/intext_numbers.do, which writes out_synth/tables/Intext_numbers.log in the public workflow and out/tables/Intext_numbers.log in the confidential workflow. The program contains only calculations for numerical statements derived from package data. Contextual facts supported by external citations and author-chosen specification values are not included in the program or log.
8. Generate the main-paper and appendix exhibits that do not require completed Fortran model results or the Figure B9 R helper.

Subsequently, run the Figure B9 R helper and the 30 Fortran model scenarios.

Lastly, code/produce_exhibits.do produces Figure 4, Figure B9, and the remaining generated exhibits; Table B6 is the manual exception documented in the exhibit map.

Instructions to Replicators

1. Set Up the Folder

Clone or unpack the replication package and start from the repository root.

```
cd replication
mkdir -p intermediate_data model_data simulation_data \
  out_synth/tables out_synth/figures
```

The public archive should contain synthetic_data/raw/, real_data/, data/public/, and code/. The synthetic raw data are already included; the generator is included only so the Data Editor can inspect or regenerate them if needed.

2. Confirm the Stata Working Directory

For the public synthetic-data replication, use code/master_synth.do. No path edits are required. Start Stata in the repository root. The drivers set \$rootpath from Stata's current working directory and verify the expected driver path before any build code runs. If Stata did not start there, change directories once:

```
cd ".../replication"
```

The drivers also accept being launched from the repository's code/ directory and automatically return to the repository root. Do not change the relative structure under synthetic_data/raw/.

The public driver then defines the remaining globals from the detected root:

```
global code = "$rootpath/code/"
global confdata = "$rootpath/synthetic_data"
global raw = "$rootpath/real_data"
global intdata = "$rootpath/intermediate_data"
global est = "$rootpath/model_data/"
global sim = "$rootpath/simulation_data/"
global out = "$rootpath/out_synth/"
```

For a confidential-data run, use code/master.do and point \$confdata to the approved local WDPI data location. Confidential data are not part of the public archive.

3. Confirm or Regenerate Synthetic Input Data

The synthetic public package should include the fictitious staff and student files listed in the “Synthetic Public Data” section above. Keep the shareable auxiliary inputs in `real_data/`.

To regenerate the simulated raw inputs from scratch, run the public simulation script from the repository root:

```
do synthetic_data/code/make_simulated_data.do
```

This recreates `synthetic_data/raw/`, refreshes `synthetic_data/documentation/synthetic_data_manifest.csv` and `synthetic_data/documentation/synthetic_data_notes.md`, and runs structural validation plus a scratch build-workflow smoke check. Run this optional regeneration only after the SSC packages checked by the master have been installed. It does not create the exhibit outputs; those are generated by the synthetic master in the next step.

4. Run the Synthetic Stata Build

From Stata:

```
do code/master_synth.do
```

From a Unix-like shell, if Stata is available on the command line:

```
stata-mp -b do code/master_synth.do
```

Replace `stata-mp` with the local Stata executable name if needed.

The synthetic run writes tables to `out_synth/tables/` and figures to `out_synth/figures/`. Immediately before the first exhibit script, the master runs `code/descriptives/intext_numbers.do` and writes `out_synth/tables/Intext_numbers.log`. The concise results-only log reports only the calculated values corresponding to statements in the paper and Online Appendix. Its first line calculates the district proficiency ranges from the included ED Data Express file. Citation-only contextual facts and author-chosen specification values are omitted. It also regenerates local working files in `intermediate_data/`, `model_data/`, and `simulation_data/`. The included `out_synth/` folder contains the expected outputs written directly by `code/master_synth.do`. Running the Figure B9 helper, the Fortran models, and `code/produce_exhibits.do` adds the remaining exhibits to the same folder. The public workflow should not write to `out/`.

5. Run R Simulation Script

Run the R code that produces the simulation underlying Figure B9. The helper locates the replication root from its own file path, so it can be invoked from the repository root without editing the script or changing the working directory. It installs missing packages in the configured R user library:

```
Rscript code/Figure_B9/cutoff_prefs_id_example_graph.R
```

6. Run Fortran Model Code

All model-estimation and counterfactual code is under `code/fortran_model/`. Shared modules are stored once in `code/fortran_model/common_f90s/`. The runner compiles the `entry/` branch for the `5_exitentry` and `6_forward` scenarios and the `non_entry/` branch for all other scenarios. Each scenario directory retains its driver, makefile, configuration, belief inputs, and expected results so that the directory maps directly to one model exercise and can be run independently.

The portable runner first verifies that `mpifort` is installed and then compiles all 30 scenarios. Compilation is the default so that invoking the script cannot accidentally begin the potentially long calculations:

```
code/fortran_model/run_all_local.sh --compile-only
```

On macOS with Homebrew, the required compiler and MPI tools can be installed with:

```
brew install gcc open-mpi openblas
```

On Debian/Ubuntu, the corresponding system packages are commonly installed with:

```
sudo apt-get install gfortran openmpi-bin libopenmpi-dev \
  libblas-dev liblapack-dev
```

After compilation, run every scenario with an explicit MPI rank count:

```
code/fortran_model/run_all_local.sh --run --ranks <ranks>
```

On the tested 128 GB Apple M4 Max machine, eight MPI ranks are recommended. The largest executable declares approximately 6.3 GB of static data per rank, so this setting provides useful parallelism while retaining substantial memory headroom. Rank counts for other systems should be chosen according to available memory.

To compile or run only one scenario, add its path relative to `code/fortran_model/`:

```
code/fortran_model/run_all_local.sh --run --ranks <ranks> \
--scenario 1_baseline
```

The runner changes into each scenario directory before execution, as required by the relative input paths in the model code. Standard output and error are written to the scenario's `result/run_main.log`; numerical result text files remain in the same `result/` directory.

Each scenario also retains its original makefile, written for a Slurm cluster. Its compile target requires `mpifort` plus BLAS and LAPACK:

```
make compile_mpi_slurm
```

The common Slurm script actually used for the submitted scenario runs is included at `code/fortran_model/example_slurm_batchfile`. It was reused across the scenarios, requested 128 MPI tasks and 200 GB of memory on partition `fu`, wrote the job log to `result/run_main.log`, set site-specific UCX/OpenMPI transport options, and launched `mpiexec run_main`. It did not specify a wall time. Another cluster user must adapt the partition, resource request, loaded modules or software environment, and transport settings. To use it with a scenario makefile, copy it into that scenario as `run_main.batch`, create the result directory, and submit through the makefile; for example, from the repository root:

```
cp code/fortran_model/example_slurm_batchfile \
code/fortran_model/1_baseline/run_main.batch
mkdir -p code/fortran_model/1_baseline/result
cd code/fortran_model/1_baseline
make run_mpi_slurm
```

Do not use these cluster-specific settings without reviewing them. Users without Slurm should use `run_all_local.sh` as described above. The scenario makefiles also contain Linux-specific library paths and compiler flags; the portable runner bypasses those settings.

Fortran instruction files

The Fortran `read_instructions` routine opens the scenario's `instruction.txt` and reads the leading numeric value or values on each line, sequentially. Text following those values is annotation for the reader, not a separate configuration input. The baseline public file `code/fortran_model/1_baseline/instruction.txt` has the following 19 records. The “full value” column refers to the matching file under `original_fortran_instruction_files/`; a dash means that the public and full values are identical.

Line	Fortran field	Public value	Full value	Purpose
1	<code>purpose</code>	10	—	Selects the program mode; 10 solves the baseline equilibrium.
2	<code>teachOff_decision</code>	1	—	Switches teacher/offer decisions on (1) or off (0), subject to purpose-specific overrides.
3	<code>wage_decision</code>	1	—	Switches wage decisions on (1) or off (0), subject to purpose-specific overrides.
4	<code>skip_first_inner_loop</code>	0	—	If 1, skips the first inner loop; 0 does not skip it.
5	<code>report_weight_factors</code>	1	—	If 1, enables the weight-factor diagnostic in the applicable first outer-loop round.
6	<code>simplex_length</code>	0.25	—	Sets the initial Nelder-Mead simplex length.
7	<code>maxiter</code>	2	1000	Sets the outer-loop/equilibrium maximum iteration count.
8	<code>outer_tol_pct</code>	0.005	—	Sets the percentage-difference convergence tolerance for outer-loop parameters.
9	<code>outer_tol_dif</code>	0.005	—	Is read for instruction-format compatibility but is not used by the shipped active execution paths.

Line	Fortran field	Public value	Full value	Purpose
10	h_maxiter	2	300	Sets the inner belief-optimization maximum iteration count.
11	h_tol_pct	0.1	—	Sets the percentage-difference convergence tolerance for belief parameters.
12	h_tol_dif	0.1	—	Is read for instruction-format compatibility but is not used by the shipped active execution paths.
13	rcost_experiment	0	—	Activates the zero-resistance-cost counterfactual when set to 1.
14	nomovecost_experiment	0	—	Activates the no-moving-cost counterfactual when set to 1.
15	NcopyT	5	—	Sets the number of teacher copies used in simulation/estimation.
16	NcopyD	3	—	Is read and used to form an unused district-copy count; it does not affect the shipped execution paths.
17	Neconomy	2	50	Sets the number of simulated economies for equilibrium/counterfactual runs.
18	starting_economy	1	—	Sets the first simulated economy, allowing earlier economies to be skipped.
19	s_vec	0.0 0.0	—	Sets the two-element subsidy vector across student types; the paper's (r0, r1) parameterization equals (s2, s1-s2).

purpose is a nonconsecutive mode code, not an ordered list of steps. The values used by the 30 shipped configurations are -2 (perturbation), -1 (standard errors), 1 (evaluate parameters and write moments), 7 (rigid wage), 10 (baseline equilibrium), and 18 (subsidized equilibrium). The source also retains several unused legacy/development branches. No purpose = 6 branch is implemented or required, so there is no missing sixth step. Purpose 7 turns teacher/offer decisions on and wage decisions off; purposes -1 and -2 turn teacher/offer decisions on and skip the inner loop.

The entry and forward-iteration readers use the same first 19-record layout; their implementation reads record 19 into svec_gradient and constructs s_vec at run time. They then read six additional records:

Additional field	Purpose
T_max	Number of model periods.
max_period_initbeliefs	Last period for which beliefs are supplied from an input file.
flag_entry	Switch controlling teacher entry.
flag_exit	Switch controlling teacher exit.
wage_elas	Wage elasticity used in the entry/forward model.
ncopy_entr	Number of entrant copies used in the simulation.

T_max is 2 in the seven 5_exitentry scenarios and 6 in the three 6_forward scenarios; these period counts are the same in the public and archived instruction files.

For non-entry scenarios, nteach_filepath.inc supplies the compile-time teacher count, the validation flag, and the repository-relative input path. Shared starting files are named initial_param_guess*.txt and initial_theta*.txt under common_f90s/. Entry and forward scenarios store period-specific belief inputs locally as mainbelief_50.txt, wagebelief_50.txt, and variants such as _t2, _t3, and subsequent periods.

In total, there are 30 scenario directories, each to be run in order to produce all modeling results. They are separated out and pre-set for easy replication. The scenarios are independent: none requires the generated results from another scenario as an input.

For a faster public synthetic-data replication, 25 active instruction.txt files set maxiter, h_maxiter, and Neconomy to 2. Five scenarios retain the full settings of 1000, 300, and 50: 2_rigid, estimation_result, perturbation, stderr, and validation_result. These five full settings are intentional for the public run and should not be reduced.

The original full-run configuration files are distributed under `original_fortran_instruction_files/`. All 30 archived files use `maxiter = 1000`, `h_maxiter = 300`, and `Neconomy = 50`; their relative paths match the runnable scenarios. Exactly 25 public files differ from their archived counterparts, and only those three computational-size fields differ. To run a full configuration, copy its archived file over the corresponding public testing configuration before invoking the runner with `--run`. For example:

```
cp original_fortran_instruction_files/1_baseline/instruction.txt \
  code/fortran_model/1_baseline/instruction.txt
bash code/fortran_model/run_all_local.sh --run --ranks 8 \
  --scenario 1_baseline
```

The runner does not select the full configurations automatically. Because they require many more iterations and simulated economies, their execution time is substantially longer than the reduced-run estimates reported above. Use a fresh package copy to return to the reduced public configurations after running a full configuration.

The model drivers initialize the random-number generator with the same fixed seed values. Different compilers, BLAS/MPI implementations, processor counts, or reduction orders can nevertheless produce last-digit floating-point differences; fixed seeds do not imply byte-for-byte equality across computing environments. The distributed Fortran `result/*.txt` files are based on the public synthetic workflow or have otherwise been cleared for public distribution.

7. Run Remaining Exhibits

While `master.do/master_synth.do` produce some exhibits, many exhibits depend on the Fortran results and are produced by another Stata file. Once all of the above results have been run, `code/produce_exhibits.do` can be run in Stata to complete the full suite of exhibits.

The Figure B10 map script leaves Stata in the `shapefile` directory. Before running the final driver, return to the replication root:

```
cd ".../replication"
do code/produce_exhibits.do
```

Exhibit Map

The entries below report the code and outputs found in this working tree. Output paths refer to the public simulated-data run under `out_synth/`. The confidential master writes its direct outputs to `out/`. The supplied `produce_exhibits.do` is configured for `out_synth/`; it must be pointed to `out/` before generating the corresponding restricted-data outputs. The Data Editor will not generate restricted-data outputs.

In-Text Numbers

- Code: `code/descriptives/intext_numbers.do` (calculation-based numerical statements only; citation-only contextual facts and author-chosen specification values are omitted).
- Main input for item 1: `data/public/proficiency_data/SY2021_FS175_DG583_LEA_data_files/SY2021_FS175_DG583_LEA.csv`
- Item 8 (173 of 411 public elementary districts with one school) is a descriptive calculation from the 2014 teacher panel. The program applies the elementary-teacher model-sample restrictions, counts distinct schools within each district, limits the result to the 411 model-estimation districts, and reports the resulting count. It does not require re-estimating the auxiliary models.
- Output: `out_synth/tables/Intext_numbers.log` for the public workflow and `out/tables/Intext_numbers.log` for the confidential workflow.

Main Paper Tables

Table 1: Teacher and District Characteristics (2014)

- Code/status: `code/descriptives/summary.do`
- Main input datasets: `model_data/experiencecont.dta`, `model_data/master.dta`, `model_data/contributions.dta`, `model_data/lambda.dta`, `model_data/urban.dta`, `model_data/suburban.dta`, `model_data/incumbent.dta`, `model_data/budget.dta`
- Output path: `out_synth/tables/Table_1A.log`; `out_synth/tables/Table_1B.log`

Table 2: OLS of Teacher-District Match (2014)

- Code/status: `code/auxiliary_regressions/aux_model_teachers_wages.do`
- Main input datasets: `model_data/multinomial.dta`, `model_data/matched_movers.dta`, `real_data/district_distance_sdid.dta`, `real_data/cesa.dta`
- Output path: `out_synth/tables/Table_2.log`; also writes `model_data/betaT1_JPE.csv`, `model_data/betaT1_var_JPE.csv`, `model_data/betaT2_JPE.csv`, `model_data/betaT2_var_JPE.csv`

Table 3: District Wage Schedules (2014)

- Code/status: code/descriptives/summary.do
- Main input datasets: model_data/contributions.dta, model_data/experiencecont.dta, model_data/master.dta, model_data/incumbent.dta, model_data/urban.dta, model_data/suburban.dta, model_data/lambda.dta, model_data/budget.dta, model_data/alphagrid.dta, model_data/multinomial.dta; real_data/district_county_cz.dta
- Output path: out_synth/tables/Table_3A.log; out_synth/tables/Table_3B.log; out_synth/tables/Table_3C.log

Table 4: Explaining District Reward for Teacher Contribution omega_d2 (2014)

- Code/status: code/descriptives/omega_analysis.do
- Main input datasets: intermediate_data/omega_analysis_data_JPE.dta
- Output path: out_synth/tables/Table_4.log
- Upstream model-moment outputs: code/auxiliary_regressions/aux_model_teachers_wages.do writes model_data/omega1_JPE.csv, model_data/omega1_var_JPE.csv, model_data/omega2_JPE.csv, model_data/omega2_var_JPE.csv, model_data/omega_moments_JPE.csv

Table 5: Parameter Estimates

- Code/status: code/descriptives/Table_5_Table_A1.do
- Main input datasets: code/fortran_model/common_f90s/initial_param_guess.txt, code/fortran_model/stderr/result/se_param.txt
- Output path: out_synth/tables/Table_5_Table_A1.csv

Table 6: Flexible Pay vs Rigid Pay

- Code/status: code/descriptives/Table_6.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/2_rigid/result
- Output path: out_synth/tables/Table_6.csv

Table 7: State-Funded Bonuses

- Code/status: code/descriptives/Table_7.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/3_basicCfs/new1/result, code/fortran_model/3_basicCfs/new2/result, code/fortran_model/3_basicCfs/new3/result
- Output path: out_synth/tables/Table_7.csv

Table 8: Moving Costs, Resistance Costs, and Policy Impacts

- Code/status: code/descriptives/Table_8.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/4_nomovecost/case1/result, code/fortran_model/4_nomovecost/case2/result, code/fortran_model/4_nomovecost/case2_new2/result
- Output path: out_synth/tables/Table_8.csv

Main Paper Figures

Figure 1: Variation in Teacher Salaries

- Code/status: code/descriptives/data_patterns.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/indtest_math_split_JPE.dta
- Output path: out_synth/figures/Figure_1.png

Figure 2: Rates of Teacher Movements Across Districts

- Code/status: code/descriptives/data_patterns.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta
- Output path: out_synth/figures/Figure_2a.png; out_synth/figures/Figure_2b.png

Figure 3: Variation in Teacher Salaries (sorting correlations)

- Code/status: code/descriptives/data_patterns.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/budget_overtime_JPE.dta

- Output path: out_synth/figures/Figure_3.png

Figure 4: Cumulative Distribution Function of ω_2 by Groups of λ_d

- Code/status: code/descriptives/Figure_4.do
- Main input datasets: model_data/lambda.csv, model_data/alpha1.csv, model_data/alpha2.csv
- Output path: out_synth/figures/Figure_4_omega2_cdf.png

Paper Appendix A Tables

Table A1: Other Parameter Estimates: Wage Setting Cost $R(\omega, \text{dem}_d)$

- Code/status: code/descriptives/Table_5_Table_A1.do
- Main input datasets: code/fortran_model/common_f90s/initial_param_guess.txt, code/fortran_model/stderr/result/se_param.txt
- Output path: out_synth/tables/Table_5_Table_A1.csv

Table A2: Model Fit: OLS of Teacher-District Match (post-Act 10)

- Code/status: code/descriptives/Table_A2.do
- Main input datasets: code/fortran_model/estimation_result/result/aux_and_omega_params.txt
- Output path: out_synth/tables/Table_A2.csv
- Synthetic-data note: the reduced public model makes the inferred-model regression numerically singular, so that column is missing in the distributed reference output. The remaining columns are populated.

Table A3: Model Fit: District Wage Schedules

- Code/status: code/descriptives/Table_A3_PanelA.do, code/descriptives/Table_A3_PanelB.do
- Main input datasets: code/fortran_model/estimation_result/result/aux_and_omega_params.txt, code/fortran_model/estimation_result/result/omega2_probs.txt, model_data/multinomial.dta
- Output path: out_synth/tables/Table_A3_PanelA.csv, out_synth/tables/Table_A3_PanelB.csv

Online Appendix B Tables

Table B1: Estimated parameters of teacher effectiveness

- Code/status: code/build/valueadded_calculation.do
- Main input datasets: intermediate_data/teachers_panel_nova_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta
- Output path: out_synth/tables/Table_B1.log; also writes intermediate_data/va_highlow_math_expdisc_JPE.dta

Table B2: Distribution of teacher effectiveness

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: model_data/contributions.dta
- Output path: out_synth/tables/Table_B2.log

Table B3: Test for Forecast Unbiasedness and Non-Random Teacher Sorting

- Code/status: code/descriptives/rothstein_test.do
- Main input datasets: intermediate_data/indtest_math_split_JPE.dta, intermediate_data/teachers_panel_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta
- Output path: out_synth/tables/Table_B3.tex

Table B4: Correlation of Teacher Effectiveness between Model 1 and Model 2

- Code/status: code/build/valueadded_calculation_classmodel.do
- Main input datasets: intermediate_data/teachers_panel_nova_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/va_highlow_math_expdisc_JPE.dta
- Output path: out_synth/tables/Table_B4.log; also writes intermediate_data/va_classmodel_JPE.dta

Table B5: Auxiliary Models Aux 1a and Aux 1b, Under Achievement Models 1 and 2

- Code/status: code/auxiliary_regressions/auxmodels_teachers_varobust.do

- Main input datasets: intermediate_data/va_classmodel_JPE.dta, model_data/multinomial.dta, real_data/district_distance_sdid.dta, real_data/cesa.dta
- Output path: out_synth/tables/Table_B5.log; also writes model_data/multinomial_varobust_JPE.dta, model_data/matched_movers_varobust_JPE.dta

Table B6: Comparison of Signal-to-Noise Ratios with Estimates of Math Teacher Value-Added in the Literature

- Code/status: No standalone replication script. The table is generated using estimates from the literature.
- Main inputs: Author calculations for the paper's value-added measures and comparisons reported in the cited literature; no separate input dataset is read by the replication workflow.
- Output path: No standalone generated file; the table is rendered as part of the online appendix.

Table B7: Sum of Squared Test Score Residuals Under c and Under (c1, c2)

- Code/status: code/descriptives/c1c2_variation_studentlevel.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta, intermediate_data/va_nocomparativeadv_JPE.dta, intermediate_data/indtest_math_split_JPE.dta
- Output path: out_synth/tables/Table_B7.log

Table B8: Achievement Production Function: Controlling for Teachers and Students' Race/Ethnicity

- Code/status: code/descriptives/achievement_race.do
- Main input datasets: intermediate_data/teachers_new_JPE.dta, intermediate_data/teachers_panel_nova_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/va_teacherside_math_JPE.dta
- Output path: out_synth/tables/Table_B8.tex

Table B9: Cross-District Summary of Pre-Reform Wage Schedules

- Code/status: code/descriptives/schedule_summary.do
- Main input datasets: model_data/lambda.dta, intermediate_data/schedule_new_JPE.dta
- Output path: out_synth/tables/Table_B9.log

Table B10: Variation in Salaries Across and Within Districts, 2013-2016

- Code/status: code/descriptives/salary_variation.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta
- Output path: out_synth/tables/Table_B10.log

Table B11: Estimates of Aux 1a Assuming Noisy Measures of (c1, c2)

- Code/status: code/auxiliary_regressions/auxiliary_models_teachers_info.do
- Main input datasets: model_data/matched_movers.dta, model_data/multinomial.dta, model_data/matched_movers_info0-3.dta, model_data/multinomial_info0-3.dta
- Output path: out_synth/tables/Table_B11.txt

Table B12: OLS of Teacher-District Matches (Aux 1a): Baseline and Excluding Bottom-Tail Matches

- Code/status: code/auxiliary_regressions/auxiliary_models_teachers_nobottom.do
- Main input datasets: model_data/matched_movers.dta, model_data/matched_movers_nobottom.dta, model_data/multinomial.dta, real_data/district_distance_sdid.dta
- Output path: out_synth/tables/Table_B12.tex

Table B13: Teachers With One vs Many Offers: Comparison

- Code/status: code/auxiliary_regressions/auxiliary_models_teachers_richprefs.do
- Main input datasets: model_data/matched_movers.dta, model_data/multinomial.dta, real_data/district_distance_sdid.dta
- Output path: out_synth/tables/Table_B13.tex

Table B14: Parameter Permutation Exercise: Change in Estimates of Auxiliary Models from Parameter Perturbation

- Code/status: code/descriptives/Table_B14.do
- Main input datasets: code/fortran_model/perturbation/result/perturb_aux_full.txt, code/fortran_model/perturbation/result/weight_mat.txt

- Output path: out_synth/tables/Table_B14.csv

Table B15: Allocating Teachers at Will

- Code/status: code/descriptives/Table_B15.do
- Main input datasets: model_data/multinomial.dta
- Output path: out_synth/tables/Table_B15.csv

Table B16: State-Funded Teacher Bonuses: Extensive Margin

- Code/status: code/descriptives/Table_B16.do
- Main input datasets: code/fortran_model/5_exitentry/baseline/result, code/fortran_model/5_exitentry/new1_05/result, code/fortran_model/5_exitentry/new1_1/result, code/fortran_model/5_exitentry/new1_15/result, code/fortran_model/5_exitentry/new2_05/result, code/fortran_model/5_exitentry/new2_1/result, code/fortran_model/5_exitentry/new2_15/result
- Output path: out_synth/tables/Table_B16.csv

Table B17: State-Funded Teacher Bonuses: Extensive Margin and Repeated Game

- Code/status: code/descriptives/Table_B17.do
- Main input datasets: code/fortran_model/6_forward/forward_baseline/result, code/fortran_model/6_forward/forward_new1/result, code/fortran_model/6_forward/forward_new2/result
- Period rule: period 6, the terminal period in all three forward scenarios.
- Output path: out_synth/tables/Table_B17.csv

Table B18: State-Funded Bonuses

- Code/status: code/descriptives/Table_B18.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/3_basicCfs/new1/result, code/fortran_model/3_basicCfs/new2/result, code/fortran_model/3_basicCfs/new3/result, code/fortran_model/3_basicCfs/new4/result
- Output path: out_synth/tables/Table_B18.csv

Table B19: Program Effects With Higher Bonus Rates

- Code/status: code/descriptives/Table_B19.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/3_basicCfs/new1/result, code/fortran_model/7_higherbonus/new1_15/result, code/fortran_model/7_higherbonus/new1_2/result, code/fortran_model/7_higherbonus/new1_25/result
- Output path: out_synth/tables/Table_B19.csv

Table B20: Program Effects with $R(\cdot) = 0$

- Code/status: code/descriptives/Table_B20.do
- Main input datasets: code/fortran_model/1_baseline/result, code/fortran_model/8_noRcost/new1/result, code/fortran_model/8_noRcost/new2/result, code/fortran_model/8_noRcost/new3/result, code/fortran_model/8_noRcost/new4/result
- Output path: out_synth/tables/Table_B20.csv

Table B21: Teacher and District Characteristics (2010)

- Code/status: code/descriptives/summary_simulation.do
- Main input datasets: simulation_data/experiencecont_pre_NEW.dta, simulation_data/master_pre_NEW.dta, simulation_data/contributions_pre_NEW.dta, simulation_data/lambda_pre_NEW.dta, simulation_data/urban_pre_NEW.dta, simulation_data/suburban_pre_NEW.dta, simulation_data/incumbent_pre_NEW.dta, simulation_data/budget_NEW.dta
- Output path: out_synth/tables/Table_B21A.log; out_synth/tables/Table_B21B.log

Table B22: Model Fit: Average District Employee Characteristics

- Code/status: code/descriptives/Table_B22.do
- Main input datasets: code/fortran_model/estimation_result/result/other_aux_params.txt
- Output path: out_synth/tables/Table_B22.csv

Table B23: Model Fit: OLS of District Wage Schedule

- Code/status: code/descriptives/Table_B23.do
- Main input datasets: code/fortran_model/estimation_result/result/aux_and_omega_params.txt
- Output path: out_synth/tables/Table_B23.csv

Table B24: Model Validation: OLS of Teacher-District Match (pre-Act 10)

- Code/status: code/descriptives/Table_B24.do
- Main input datasets: code/fortran_model/validation_result/result/aux_and_omega_params.txt
- Output path: out_synth/tables/Table_B24.csv
- Synthetic-data note: the reduced public data do not identify these validation regressions, so the numeric cells are missing in the distributed reference output. This is expected for the fictitious data and does not occur in the confidential-data estimates reported in the paper.

Table B25: Model Validation: Average District Employee Characteristics (pre-Act 10)

- Code/status: code/descriptives/Table_B25.do
- Main input datasets: code/fortran_model/validation_result/result/other_aux_params.txt
- Output path: out_synth/tables/Table_B25.csv

Online Appendix B Figures

Figure B1: Distribution of Teacher Effectiveness

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: model_data/contributions.dta
- Output path: out_synth/figures/Figure_B1.png

Figure B2: Relationship Between c_1 and c_2

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: model_data/multinomial.dta
- Output path: out_synth/figures/Figure_B2.png

Figure B3: Relationship Between $W_d^0(x_i)$ and w_{it}^0

- Code/status: code/descriptives/appendix_figures.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/schedule_new_JPE.dta
- Output path: out_synth/figures/Figure_B3.png

Figure B4: Relationship Between Wage Deviations Under Two- and Three-Omega Rules

- Code/status: code/descriptives/gen_omegas_3omegas.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/schedule_new_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/alphagrid_wages_8groups_JPE.dta
- Output path: out_synth/figures/Figure_B4.png; also writes intermediate_data/alphagrid_wages_8groups_3omegas_JPE.dta

Figure B5: Relationship Between Wage Deviations, Untenured Teachers

- Code/status: code/descriptives/gen_omegas_tenured.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/schedule_new_JPE.dta, intermediate_data/indtest_math_split_JPE.dta, intermediate_data/alphagrid_8groups_lambdad_JPE.dta, intermediate_data/alphagrid_wages_8groups_JPE.dta
- Output path: out_synth/figures/Figure_B5.png; also writes intermediate_data/alphagrid_wages_8groups_tenured_JPE.dta

Figure B6: Relationship Between Wage Deviations, No-Experience Teachers

- Code/status: code/descriptives/gen_omegas_exp1cutoff.do
- Main input datasets: intermediate_data/teachers_panel_JPE.dta, intermediate_data/schedule_new_JPE.dta, intermediate_data/indtest_math_split_JPE.dta,

- intermediate_data/alphagrid_8groups_lambda_JPE.dta,
- intermediate_data/alphagrid_wages_8groups_JPE.dta
- Output path: out_synth/figures/Figure_B6.png; also writes intermediate_data/alphagrid_wages_8groups_exp1cutoff_JPE.dta

Figure B7: Share of Teachers Who Switch In and Out of Math Teaching, By Year

- Code/status: code/descriptives/budget_shifts.do
- Main input datasets: intermediate_data/teachers_new_JPE.dta
- Output path: out_synth/figures/Figure_B7.png

Figure B8: Share of Districts' Budgets Spent on Teacher Salaries, by Grade and Subject

- Code/status: code/descriptives/budget_shifts.do
- Main input datasets: intermediate_data/teachers_new_JPE.dta
- Output path: out_synth/figures/Figure_B8A.png; out_synth/figures/Figure_B8B.png; out_synth/figures/Figure_B8C.png; out_synth/figures/Figure_B8D.png

Figure B9: Identification Illustrated

- Code/status: code/Figure_B9/Figure_B9.do; helper code/Figure_B9/cutoff_prefs_id_example_graph.R creates the input
- Main input datasets: code/Figure_B9/cutoffs_id_exercise.dta
- Output path: out_synth/figures/Figure_B9_b_02.png; out_synth/figures/Figure_B9_b_1.png; out_synth/figures/Figure_B9_b_5.png

Figure B10: Distribution of Lambda Across Wisconsin Districts

- Code/status: code/descriptives/lambda_map.do
- Main input datasets: real_data/district_shapefiles/WI_School_Districts.*, intermediate_data/indtest_math_split_JPE.dta, model_data/lambda.dta
- Output path: out_synth/figures/Figure_B10.png

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Contact for questions:

Barbara Biasi, Yale School of Management and NBER, barbara.biasi@yale.edu.